

CA-S

ASSIGNMENT 3

TRENDS AND CLOUD SERVICES

Serverless architecture, Progressive Web Apps and the integration of AI/ML in software are significant trends in cloud computing. Serverless architecture offers benefits like automatic scaling, reduced operational overhead and cost-efficiency by allowing developers to focus on code without managing infrastructure. PWA's bridge the gap between web and native apps by offering a web based experience with app like feature and offline capabilities. AI/ML is transforming software architecture by enabling tasks like predictive analysis, personalized user experiences, and intelligent automation.

Progressive Web Apps are web applications that are designed to function both a website and a mobile app. They utilize service workers, manifest files and push notifications to provide features like offline access, installation on a device's home screen and the ability to deliver updates without requiring users to visit the website.

AI and ML are increasingly integrated into software architecture to enhance various aspects of applications. For example, machine learning can be used for tasks like natural language processing to improve user interfaces, computer vision to enhance image recognition and predictive analysis to optimize system performance. This allows developers to build more intelligent, adaptable and user-friendly applications that can learn from data and provide more personalized experiences.

Cloud computing service models including Software as a Service (SaaS), Platform as a Service (PaaS) and Infrastructure as a Service (IaaS) offer varying levels of control and responsibility. SaaS provides complete end-to-end infrastructure or the software, PaaS provides a platform for developers to build and deploy applications and IaaS offers the most basic level of control.

SaaS applications include email services

PaaS applications include AWS

IaaS applications include Microsoft Azure